

Here's a list of all the tables and their columns, along with how they are related:

Tables and Columns:

- **bank_customer**
 - `customer_id` (Primary Key)
 - `customer_name`
- **Bank_Account_Details**
 - `account_number` (Primary Key)
 - `customer_id` (Foreign Key)
 - `account_type`
 - `balance_amount`
 - `Relationship_type`
- **Bank_Account_transaction**
 - `transaction_id` (Primary Key)
 - `account_number` (Foreign Key)
 - `transaction_amount`
 - `Transaction_Date`
- **BANK_CUSTOMER_MESSAGES**
 - `message_id` (Primary Key)
 - `Event`
 - `customer_message`
- **bank_account_relationship_details**
 - `relationship_id` (Primary Key)
 - `account_number` (Foreign Key)
 - `linking_account_number` (Foreign Key)
 - `account_type`

Table Relationships:

- **bank_customer and Bank_Account_Details:**
 - `bank_customer.customer_id` (Primary Key) is related to `Bank_Account_Details.customer_id` (Foreign Key). This means a customer can have multiple bank accounts.
- **Bank_Account_Details and Bank_Account_transaction:**
 - `Bank_Account_Details.account_number` (Primary Key) is related to `Bank_Account_transaction.account_number` (Foreign Key). This indicates that each bank account can have multiple transactions.

- **Bank_Account_Details and bank_account_relationship_details:**
 - **Bank_Account_Details.account_number** (Primary Key) is related to **bank_account_relationship_details.account_number** (Foreign Key) and **bank_account_relationship_details.linking_account_number** (Foreign Key). This table establishes relationships between different bank accounts, where one account can be linked to another (e.g., a savings account linked to a credit card).
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1. Table Creation (DDL)

-- Customers

```
CREATE TABLE bank_customer (  
    customer_id INT PRIMARY KEY,  
    customer_name VARCHAR(100)  
);
```

-- Accounts

```
CREATE TABLE Bank_Account_Details (  
    account_number INT PRIMARY KEY,  
    customer_id INT,  
    account_type VARCHAR(50),  
    balance_amount DECIMAL(12,2),  
    relationship_type VARCHAR(50),  
    FOREIGN KEY (customer_id) REFERENCES bank_customer(customer_id)  
);
```

-- Transactions

```
CREATE TABLE Bank_Account_transaction (  
    transaction_id INT PRIMARY KEY,  
    account_number INT,  
    transaction_amount DECIMAL(12,2),  
    transaction_date DATE,  
    FOREIGN KEY (account_number) REFERENCES Bank_Account_Details(account_number)  
);
```

-- Messages

```
CREATE TABLE BANK_CUSTOMER_MESSAGES (  
    message_id INT PRIMARY KEY,  
    event VARCHAR(100),  
    customer_message VARCHAR(255)
```

```
);
```

```
-- Account Relationships
```

```
CREATE TABLE bank_account_relationship_details (  
    relationship_id INT PRIMARY KEY,  
    account_number INT,  
    linking_account_number INT,  
    account_type VARCHAR(50),  
    FOREIGN KEY (account_number) REFERENCES Bank_Account_Details(account_number),  
    FOREIGN KEY (linking_account_number) REFERENCES  
Bank_Account_Details(account_number)  
);
```

2. Sample Data Inserts

bank_customer

```
INSERT INTO bank_customer (customer_id, customer_name) VALUES  
(1, 'Customer_1'), (2, 'Customer_2'), (3, 'Customer_3'), (4, 'Customer_4'), (5, 'Customer_5'),  
(6, 'Customer_6'), (7, 'Customer_7'), (8, 'Customer_8'), (9, 'Customer_9'), (10, 'Customer_10'),  
(11, 'Customer_11'), (12, 'Customer_12'), (13, 'Customer_13'), (14, 'Customer_14'), (15,  
'Customer_15'),  
(16, 'Customer_16'), (17, 'Customer_17'), (18, 'Customer_18'), (19, 'Customer_19'), (20,  
'Customer_20'),  
(21, 'Customer_21'), (22, 'Customer_22'), (23, 'Customer_23'), (24, 'Customer_24'), (25,  
'Customer_25'),  
(26, 'Customer_26'), (27, 'Customer_27'), (28, 'Customer_28'), (29, 'Customer_29'), (30,  
'Customer_30'),  
(31, 'Customer_31'), (32, 'Customer_32'), (33, 'Customer_33'), (34, 'Customer_34'), (35,  
'Customer_35'),  
(36, 'Customer_36'), (37, 'Customer_37'), (38, 'Customer_38'), (39, 'Customer_39'), (40,  
'Customer_40'),  
(41, 'Customer_41'), (42, 'Customer_42'), (43, 'Customer_43'), (44, 'Customer_44'), (45,  
'Customer_45'),  
(46, 'Customer_46'), (47, 'Customer_47'), (48, 'Customer_48'), (49, 'Customer_49'), (50,  
'Customer_50');
```

Bank_Account_Details

INSERT INTO Bank_Account_Details (account_number, customer_id,
account_type, balance_amount, relationship_type) VALUES

(1001, 1, 'Savings', 5000.00, 'Primary'),
(1002, 2, 'Current', 12000.00, 'Primary'),
(1003, 3, 'Loan', -20000.00, 'Primary'),
(1004, 4, 'Savings', 8000.00, 'Primary'),
(1005, 5, 'Credit Card', -5000.00, 'Primary'),
(1006, 6, 'Savings', 6000.00, 'Primary'),
(1007, 7, 'Current', 15000.00, 'Primary'),
(1008, 8, 'Loan', -10000.00, 'Primary'),
(1009, 9, 'Savings', 7000.00, 'Primary'),
(1010, 10, 'Credit Card', -2000.00, 'Primary'),
(1011, 11, 'Savings', 5500.00, 'Primary'),
(1012, 12, 'Current', 14000.00, 'Primary'),
(1013, 13, 'Loan', -8000.00, 'Primary'),
(1014, 14, 'Savings', 9000.00, 'Primary'),
(1015, 15, 'Credit Card', -3000.00, 'Primary'),
(1016, 16, 'Savings', 6200.00, 'Primary'),
(1017, 17, 'Current', 10000.00, 'Primary'),
(1018, 18, 'Loan', -12000.00, 'Primary'),
(1019, 19, 'Savings', 4800.00, 'Primary'),
(1020, 20, 'Credit Card', -1500.00, 'Primary'),
(1021, 21, 'Savings', 5000.00, 'Primary'),
(1022, 22, 'Current', 13000.00, 'Primary'),
(1023, 23, 'Loan', -25000.00, 'Primary'),
(1024, 24, 'Savings', 7200.00, 'Primary'),
(1025, 25, 'Credit Card', -4000.00, 'Primary'),
(1026, 26, 'Savings', 8000.00, 'Primary'),
(1027, 27, 'Current', 16000.00, 'Primary'),
(1028, 28, 'Loan', -9000.00, 'Primary'),
(1029, 29, 'Savings', 6500.00, 'Primary'),
(1030, 30, 'Credit Card', -2200.00, 'Primary'),
(1031, 31, 'Savings', 7000.00, 'Primary'),
(1032, 32, 'Current', 17000.00, 'Primary'),
(1033, 33, 'Loan', -15000.00, 'Primary'),
(1034, 34, 'Savings', 9400.00, 'Primary'),
(1035, 35, 'Credit Card', -2800.00, 'Primary'),
(1036, 36, 'Savings', 7700.00, 'Primary'),
(1037, 37, 'Current', 18000.00, 'Primary'),
(1038, 38, 'Loan', -7000.00, 'Primary'),
(1039, 39, 'Savings', 6800.00, 'Primary'),
(1040, 40, 'Credit Card', -1900.00, 'Primary'),
(1041, 41, 'Savings', 5600.00, 'Primary'),

(1042, 42, 'Current', 13500.00, 'Primary'),
(1043, 43, 'Loan', -11000.00, 'Primary'),
(1044, 44, 'Savings', 8800.00, 'Primary'),
(1045, 45, 'Credit Card', -3200.00, 'Primary'),
(1046, 46, 'Savings', 6000.00, 'Primary'),
(1047, 47, 'Current', 15500.00, 'Primary'),
(1048, 48, 'Loan', -13000.00, 'Primary'),
(1049, 49, 'Savings', 7200.00, 'Primary'),
(1050, 50, 'Credit Card', -2100.00, 'Primary');

Bank_Account_transaction

INSERT INTO Bank_Account_transaction (transaction_id, account_number,
transaction_amount, transaction_date) VALUES
(2001, 1001, -500.00, '2025-09-01'),
(2002, 1002, 2000.00, '2025-09-01'),
(2003, 1003, -1500.00, '2025-09-02'),
(2004, 1004, 1000.00, '2025-09-02'),
(2005, 1005, -700.00, '2025-09-03'),
(2006, 1006, 3000.00, '2025-09-03'),
(2007, 1007, -200.00, '2025-09-04'),
(2008, 1008, -1200.00, '2025-09-04'),
(2009, 1009, 4000.00, '2025-09-05'),
(2010, 1010, -900.00, '2025-09-05'),
(2011, 1011, -600.00, '2025-09-06'),
(2012, 1012, 2500.00, '2025-09-06'),
(2013, 1013, -1800.00, '2025-09-07'),
(2014, 1014, 3200.00, '2025-09-07'),
(2015, 1015, -700.00, '2025-09-08'),
(2016, 1016, 2800.00, '2025-09-08'),
(2017, 1017, -500.00, '2025-09-09'),
(2018, 1018, -2000.00, '2025-09-09'),
(2019, 1019, 3500.00, '2025-09-10'),
(2020, 1020, -800.00, '2025-09-10'),
(2021, 1021, -600.00, '2025-09-11'),
(2022, 1022, 2300.00, '2025-09-11'),
(2023, 1023, -2100.00, '2025-09-12'),
(2024, 1024, 2900.00, '2025-09-12'),
(2025, 1025, -500.00, '2025-09-13'),
(2026, 1026, 3100.00, '2025-09-13'),
(2027, 1027, -400.00, '2025-09-14'),
(2028, 1028, -2500.00, '2025-09-14'),
(2029, 1029, 2700.00, '2025-09-15'),

(2030, 1030, -600.00, '2025-09-15'),
(2031, 1031, -700.00, '2025-09-16'),
(2032, 1032, 3400.00, '2025-09-16'),
(2033, 1033, -2200.00, '2025-09-17'),
(2034, 1034, 2800.00, '2025-09-17'),
(2035, 1035, -800.00, '2025-09-18'),
(2036, 1036, 2600.00, '2025-09-18'),
(2037, 1037, -500.00, '2025-09-19'),
(2038, 1038, -1700.00, '2025-09-19'),
(2039, 1039, 2900.00, '2025-09-20'),
(2040, 1040, -700.00, '2025-09-20'),
(2041, 1041, -900.00, '2025-09-21'),
(2042, 1042, 3100.00, '2025-09-21'),
(2043, 1043, -1900.00, '2025-09-22'),
(2044, 1044, 2700.00, '2025-09-22'),
(2045, 1045, -800.00, '2025-09-23'),
(2046, 1046, 2500.00, '2025-09-23'),
(2047, 1047, -600.00, '2025-09-24'),
(2048, 1048, -1300.00, '2025-09-24'),
(2049, 1049, 2400.00, '2025-09-25'),
(2050, 1050, -700.00, '2025-09-25');

BANK_CUSTOMER_MESSAGES

INSERT INTO BANK_CUSTOMER_MESSAGES (message_id, event, customer_message)
VALUES

(3001, 'Low Balance Alert', 'Your account balance is below the minimum requirement.'),
(3002, 'Transaction Alert', 'You have spent INR 500 on your Credit Card.'),
(3003, 'Loan EMI Reminder', 'Your loan EMI of INR 1500 is due tomorrow.'),
(3004, 'Welcome Message', 'Thank you for opening an account with us.'),
(3005, 'Credit Card Alert', 'Your credit card limit is about to be reached.'),
(3006, 'Deposit Confirmation', 'Your deposit of INR 2000 has been credited.'),
(3007, 'Withdrawal Alert', 'You withdrew INR 500 from your account.'),
(3008, 'Overdue Loan Alert', 'Your loan payment is overdue.'),
(3009, 'Salary Credit', 'Your salary has been credited successfully.'),
(3010, 'Statement Ready', 'Your monthly bank statement is available.'),
(3011, 'Promo Offer', 'Get 10% cashback on shopping.'),
(3012, 'Debit Card Alert', 'Your debit card transaction is successful.'),
(3013, 'Account Upgrade', 'Your account has been upgraded.'),
(3014, 'FD Renewal', 'Your fixed deposit is due for renewal.'),
(3015, 'OTP Sent', 'An OTP has been sent to your registered number.'),
(3016, 'Cheque Cleared', 'Your cheque has been cleared.'),
(3017, 'Cheque Bounced', 'Your cheque has bounced.'),

(3018, 'KYC Update', 'Please update your KYC documents.'),
(3019, 'Loan Approved', 'Your loan has been approved.'),
(3020, 'Loan Disbursed', 'Your loan amount has been credited.'),
(3021, 'Bill Payment', 'Your electricity bill has been paid.'),
(3022, 'Recharge Success', 'Your mobile recharge was successful.'),
(3023, 'Mutual Fund Alert', 'Your SIP installment is due.'),
(3024, 'Insurance Premium', 'Your insurance premium is due.'),
(3025, 'Credit Score Update', 'Your credit score has been updated.'),
(3026, 'ATM Withdrawal', 'Cash withdrawn at ATM.'),
(3027, 'POS Transaction', 'Card used at POS terminal.'),
(3028, 'Interest Credit', 'Interest credited to your account.'),
(3029, 'Penalty Charged', 'A penalty has been applied.'),
(3030, 'Reward Points', 'You earned 500 reward points.'),
(3031, 'Investment Alert', 'Your investment portfolio has grown.'),
(3032, 'Gold Loan Alert', 'Your gold loan EMI is due.'),
(3033, 'Education Loan Alert', 'Your education loan EMI is pending.'),
(3034, 'Home Loan Alert', 'Your home loan EMI is due.'),
(3035, 'Personal Loan Alert', 'Your personal loan EMI is due.'),
(3036, 'Card Blocked', 'Your card has been temporarily blocked.'),
(3037, 'Card Activated', 'Your card has been activated.'),
(3038, 'NEFT Success', 'Your NEFT transaction was successful.'),
(3039, 'RTGS Success', 'Your RTGS transaction was successful.'),
(3040, 'IMPS Success', 'Your IMPS transaction was successful.'),
(3041, 'UPI Success', 'Your UPI payment is complete.'),
(3042, 'Overdraft Alert', 'You have exceeded overdraft limit.'),
(3043, 'Dormant Account', 'Your account is dormant.'),
(3044, 'Reactivation Notice', 'Your account has been reactivated.'),
(3045, 'Card Replacement', 'Your new debit card is dispatched.'),
(3046, 'Loan Prepayment', 'Your prepayment request is approved.'),
(3047, 'EMI Bounce', 'Your EMI transaction failed.'),
(3048, 'EMI Deducted', 'Your EMI has been deducted.'),
(3049, 'Account Closure', 'Your account has been closed.'),
(3050, 'System Maintenance', 'Banking system will be under maintenance.');

bank_account_relationship_details

```
INSERT INTO bank_account_relationship_details (relationship_id, account_number,  
linking_account_number, account_type) VALUES  
(4001, 1001, 1005, 'Savings to Credit Card'),  
(4002, 1002, 1003, 'Current to Loan'),  
(4003, 1004, 1006, 'Savings to Savings'),  
(4004, 1007, 1008, 'Current to Loan'),  
(4005, 1009, 1010, 'Savings to Credit Card'),
```

(4006, 1011, 1013, 'Savings to Loan'),
(4007, 1012, 1014, 'Current to Savings'),
(4008, 1015, 1016, 'Credit Card to Savings'),
(4009, 1017, 1018, 'Current to Loan'),
(4010, 1019, 1020, 'Savings to Credit Card'),
(4011, 1021, 1023, 'Savings to Loan'),
(4012, 1022, 1024, 'Current to Savings'),
(4013, 1025, 1026, 'Credit Card to Savings'),
(4014, 102

7, 1028, 'Current to Loan'),
(4015, 1029, 1030, 'Savings to Credit Card'),
(4016, 1031, 1033, 'Savings to Loan'),
(4017, 1032, 1034, 'Current to Savings'),
(4018, 1035, 1036, 'Credit Card to Savings'),
(4019, 1037, 1038, 'Current to Loan'),
(4020, 1039, 1040, 'Savings to Credit Card'),
(4021, 1041, 1043, 'Savings to Loan'),
(4022, 1042, 1044, 'Current to Savings'),
(4023, 1045, 1046, 'Credit Card to Savings'),
(4024, 1047, 1048, 'Current to Loan'),
(4025, 1049, 1050, 'Savings to Credit Card'),
(4026, 1001, 1002, 'Savings to Current'),
(4027, 1003, 1005, 'Loan to Credit Card'),
(4028, 1006, 1007, 'Savings to Current'),
(4029, 1008, 1009, 'Loan to Savings'),
(4030, 1010, 1012, 'Credit Card to Current'),
(4031, 1013, 1014, 'Loan to Savings'),
(4032, 1015, 1017, 'Credit Card to Current'),
(4033, 1016, 1018, 'Savings to Loan'),
(4034, 1019, 1021, 'Savings to Savings'),
(4035, 1020, 1022, 'Credit Card to Current'),
(4036, 1023, 1024, 'Loan to Savings'),
(4037, 1025, 1027, 'Credit Card to Current'),
(4038, 1026, 1028, 'Savings to Loan'),
(4039, 1029, 1031, 'Savings to Savings'),
(4040, 1030, 1032, 'Credit Card to Current'),
(4041, 1033, 1034, 'Loan to Savings'),
(4042, 1035, 1037, 'Credit Card to Current'),
(4043, 1036, 1038, 'Savings to Loan'),
(4044, 1039, 1041, 'Savings to Savings'),
(4045, 1040, 1042, 'Credit Card to Current'),
(4046, 1043, 1044, 'Loan to Savings'),

(4047, 1045, 1047, 'Credit Card to Current'),
(4048, 1046, 1048, 'Savings to Loan'),
(4049, 1049, 1001, 'Savings to Savings'),
(4050, 1050, 1002, 'Credit Card to Current');